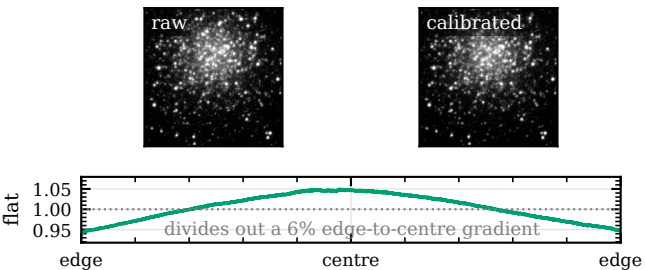
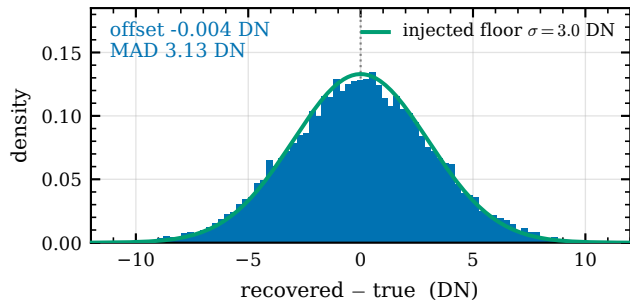


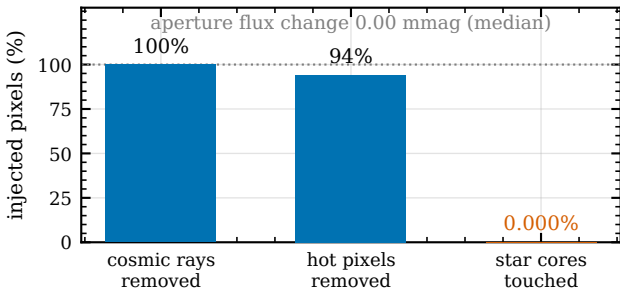
(a) vignette + fixed pattern removed



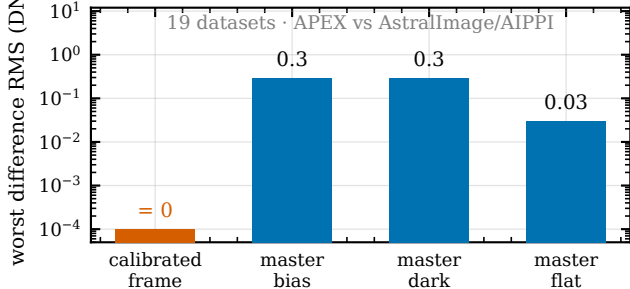
(b) truth recovery: offset-free, at the noise floor



(c) cosmetics removed, stars untouched



(d) independent engine: calibrated frames bit-identical



(a) M13 V60 s, Moravian C3-61000 (2×2) · (b) synthetic bias/dark/flat injected and inverted through APEX Step 0, seed 1234 (same fixture as the regression test)

(c) apex.benchmark.cosmetic_validate: 80 stars, 115 CR pixels, 120 hot pixels injected · (d) 19 cluster/variable datasets, 9 bands · gen: fig_calibration_step0.py